

Disclaimer

Environment: This notebook has been tested with:

- Julia v1.12.x
- JuMP.jl v1.30.x
- HiGHS v1.24.1

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Case 2: Multi-Objective Optimization

Summary of main Notations:

Notation	Description
$\mathbb{R}^n$	The decision space, with n the number of variables
$\mathbb{R}^p$	The objective space, with p the number of objectives
X	The set of feasible solutions in $\mathbb{R}^n$
Y $= z(X)$	The set of images of feasible solutions in $\mathbb{R}^p$
X_E $\subseteq X$	A complete set of efficient solutions
$Y_N \subseteq Y$	The set of nondominated points, with $Y_N = Y_{SN} \cup Y_{NN}$
Y_{SN} $\subseteq Y_N$	The set of supported nondominated points
Y_{NN} $\subseteq Y_N$	The set of nonsupported nondominated points
$y^I \in \mathbb{R}^p$	The ideal point

Notation	Description
y^U $\in \mathbb{R}^p$	The utopian point, with (in $\max$) $y^U = y^I + \epsilon$
y^N $\in \mathbb{R}^p$	The nadir point

The base bi-objective model:

Given the bi-objective 01 Unidimensional Knapsack Problem (bi-01UKP) formulated by

$$z = \max \{ (p^1 x, p^2 x) \mid wx \leq c, x \in \{0, 1\}^n \}$$

and the numerical instance corresponding to

$$n = 10$$

$$p^1 = (13, 10, 3, 16, 12, 11, 1, 9, 19, 13)$$

$$p^2 = (1, 10, 3, 13, 12, 19, 16, 13, 11, 9)$$

$$w = (4, 4, 3, 5, 5, 3, 2, 3, 5, 4)$$

$$c = 19$$

perform the following progressive tasks:

1. Using the package `JuMP`, implement an implicit formulation of the bi-01UKP and display the model.
2. Using `HiGHS` as MIP solver, the package `MultiObjectiveAlgorithms` and the algorithm `MOA.EpsilonConstraint()`, compute and display X_E and Y_N , a complete set of efficient solutions and the set of nondominated points.
3. Switch to the algorithm `MOA.Dichotomy()` without rebuilding the model, and re-optimize to get Y_{SN} , the supported nondominated points. How many points are found now? Why fewer?
4. Identify $Y_{NN} = Y_N \setminus Y_{SN}$, the nonsupported nondominated points.
5. From Y_N , compute the ideal point y^I and the nadir point y^N .
6. Perform a manual scalarization in building a single-objective JuMP model (without `MultiObjectiveAlgorithms`) that maximizes the weighted sum $0.5 p^1 x + 0.5 p^2 x$ directly with `HiGHS`. Check whether the resulting point belongs to Y_{SN} .
7. Using the package `Plots`, display Y_N in the objective space $\mathbb{R}^p$, in differentiating points belonging to Y_{SN} and Y_{NN} .

8. Extending the 01 unidimensionnal knapsack problem to the tri-objective case with the given vector coefficients, report Y_N , Y_{SE} , and Y_{NN} . Display Y_N in the objective space $\mathbb{R}^p$.

Summary of in-place model modification commands:

Reusing the same `model` object avoids unnecessary reconstruction:

Change needed	Command (no rebuild)
Change MOA algorithm	<code>set_optimizer(model, MOA.Algorithm(), new_algo)</code>
Change one objective function	<code>@objective(model, Max, [obj1, new_obj2])</code>
Add a constraint	<code>@constraint(model, name, expr)</code>
Remove a constraint	<code>delete(model, mycon) & unregister(model, :mycon)</code>
Change RHS of a constraint	<code>set_normalized_rhs(con, new_value)</code>
Change a constraint coefficient	<code>set_normalized_coefficient(con, var, new_value)</code>

Solutions:

1. using the package `JuMP`, implement an implicit formulation of the bi-01UKP and display the model.

- declare the package(s) to use:

In []:

- setup the instance to solve

In []:

- setup the formulation

In []:

2. using the package `MultiObjectiveAlgorithms`, compute and display X_E and Y_N , a complete set of efficient solutions and the set of nondominated points.

- setup the MIP solver to use

Hints: see `set_optimizer()` in the slides

In []:

- setup the algorithm to use

Hints: see `set_attribute()` in the slides

In []:

- solve the problem

In []:

- query the number of points in Y_N

Hints: see `result_count()` in the slides

In []:

- query the vector values of the 3rd point

In []:

- query the value for the first objective of the 3rd point

In []:

- display X_E and Y_N

For each result, print the efficient solutions ($x \in X_E$) and its image in objective space ($y \in Y_N$).